
INTERNSHIP PROJECT DOCUMENT

Project Title

Local Business Intelligence & Analytics Platform

Duration

1 Week

Team Structure

- **4 Students per Team**
 - Each team will work on **ONE dataset (different for each team)**
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Objective

Build a **Python-based analytics system** that helps a business:

- Load sales data from CSV
- Analyze revenue, products, and customers
- Generate insights
- Visualize trends using graphs

This project simulates a **real-world business analytics tool**.

Problem Statement

A business collects sales data but does not have a system to analyze it.

Your team must build a **menu-driven analytics platform** that:

- Reads CSV data
 - Performs analysis
 - Displays insights
 - Generates reports
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SYSTEM REQUIREMENTS

1. Data Loading

- Take file name as input
 - Load CSV file
 - Handle errors (file not found, wrong format)
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2. Data Preview

- Show:
 - Total rows & columns
 - Column names
 - First 5 records
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3. Revenue Analysis

- Total revenue
 - Average order value
 - Total transactions
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4. Product Analysis

- Top-selling products
- Least-selling products
- Category-wise sales

5. Customer Analysis

- Total unique customers
- Most active customers

6. Sales Trend Analysis

- Daily / monthly sales trends

7. Visualization

Create graphs using Matplotlib / Seaborn:

- Bar chart → Top products
- Line chart → Sales trend
- Pie chart → Category distribution

8. Report Generation

- Save summary in `.txt` or `.csv` file

9. Error Handling

Handle:

- Invalid inputs
 - Missing data
 - Empty dataset
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SYSTEM MENU (MANDATORY)

1. Load Data
2. Preview Data
3. Revenue Analysis
4. Product Analysis
5. Customer Analysis
6. Sales Trends
7. Visualize Data
8. Generate Report
9. Exit

ROLE DISTRIBUTION (COMPULSORY)

◆ Student 1: Data Engineer

- File loading
- Data cleaning
- Exception handling

◆ Student 2: Core Python Developer

- Menu system
- Revenue calculations
- Functions

◆ Student 3: Data Analyst

- Pandas analysis
 - Product & customer insights
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◆ Student 4: Visualization & Reporting

- Graphs (Matplotlib / Seaborn)
 - Report generation
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PROJECT STRUCTURE (MANDATORY)

```
data_loader.py
core_logic.py
analysis.py
visualization.py
main.py
```

✗ Single file NOT allowed

RULES

- ✗ No copying full solution
- ✗ No single-file project

☒ Each student must:

- Work on their module
 - Explain their code
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EXPECTED OUTPUT

- Working CLI system
- Clean analysis results
- Graphs generated
- Report file created

FINAL SUBMISSION

- ☒ Python code (all files)
- ☒ Dataset used
- ☒ Output graphs
- ☒ Report file
- ☒ Demo

FINAL INSTRUCTION

Your system should be **generic** — it must work for any business dataset with the same structure.

Outcome

By completing this project, you will learn:

- Real-world problem solving
 - Team collaboration
 - Data analysis using Python
 - Building modular systems
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